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Application of Full Frequency Inversion in Prediction for Complex Carbonate Tidal Channel Reservoirs

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Abstract

Carbonate tidal channel reservoirs in the Middle East present significant challenges for reservoir prediction due to their strong heterogeneity, complex pore structures, and rapid lateral thickness variation. In this study, we apply full-frequency inversion (FFI) a seismic waveform-driven, high-resolution inversion technique to enhance the prediction accuracy of thin-layered tidal channel reservoirs in a carbonate setting in northern UAE. FFI integrates seismic waveform clustering, well-log calibration, and a facies-based Bayesian framework to estimate seismic impedance and other petrophysical attributes across the full frequency spectrum.

This study applies a full-frequency inversion (FFI) technique to improve both vertical and lateral resolution in tidal channel reservoir prediction. FFI is a high-resolution seismic inversion method that extends the seismic frequency bandwidth using waveform-driven analysis. By correlating seismic waveforms with well logs, wells with similar waveform characteristics are selected as training samples. A likelihood function is then constructed to reflect both the high-frequency impedance and well-log structure. Finally, frequency-band inversion results are integrated through a shaping-weighted spectrum synthesis to produce the final FFI result.

The method improves vertical resolution to ~2 meters and lateral continuity beyond what conventional inversion techniques allow. Application results show that FFI effectively delineates the 3D architecture of overlapping tidal channels and associated thin interbeds, accurately identifying high-permeability zones critical for waterflood efficiency. The integrated inversion of impedance, resistivity, and natural gamma also supports high-resolution facies classification. This approach provides a practical and transferable framework for characterizing thin heterogeneous carbonate reservoirs elsewhere.

Introduction

Carbonate reservoirs, particularly those in the Middle East, are characterized by extreme heterogeneity, complex pore systems, and highly variable sedimentary environments. These factors often hinder accurate reservoir characterization and effective development planning. Traditional seismic inversion techniques,

including sparse-spike and model-based inversion, are constrained by limited seismic bandwidth and oversimplified assumptions about reservoir continuity. Moreover, geostatistical inversion methods, though capable of simulating high-frequency components, frequently rely heavily on prior models and dense well coverage — conditions rarely met in carbonate settings.

Tidal channel reservoirs, which are common in carbonate platforms, introduce additional complexity. Their development often involves multi-stage stacking, lateral migration, and rapid pinch-outs, leading to intricate three-dimensional geometries. In such systems, high-permeability "thief zones" can cause water breakthrough during injection operations, severely limiting sweep efficiency and reducing ultimate recovery.

To address these challenges, this study introduces a full-frequency inversion (FFI) technique that combines seismic waveform analysis, facies-controlled modeling, and intelligent spectrum synthesis. The method is applied to a complex tidal channel reservoir in the northern UAE, targeting the Cretaceous carbonate formations. The objectives are to:

- Improve vertical and lateral resolution of reservoir prediction;

- Accurately delineate the spatial architecture of tidal channels and interbeds;

- Enhance prediction of key reservoir attributes such as porosity, permeability, and resistivity;

The following sections present the geological background of the study area, a detailed explanation of the FFI methodology, case study results, and a discussion of implications for carbonate reservoir prediction.

Geological Setting and Technical Challenges

The study area is located in the northern UAE and features a typical carbonate platform setting formed during the Cretaceous. The targeted formation is approximately 50 meters thick and exhibits complex tidal channel sedimentation, characterized by net-like horizontal channel patterns and vertical stacking due to multiple sedimentary cycles.

Sedimentary microfacies in this region include lagoonal margins, inner-platform shoals, tidal flats, and backshoals, deposited under relatively low-energy conditions. The resulting reservoir heterogeneity is extreme, with a wide range of porosity types, including matrix microporosity, intergranular pores, moldic pores, and grain dissolution pores. Throat sizes are primarily micro- to meso-scale, further complicating permeability prediction.

Three main technical challenges impact reservoir characterization in this area:

- Complex pore networks and weak attribute sensitivity, quantitative relationships between porosity and permeability are weak, and seismic attributes often fail to effectively indicate sweet spots. Additionally, the development of thin high permeability "thief zones" exacerbates water breakthrough and reduces sweep efficiency.
- Rapid vertical and lateral facies variation, the stacking and migration of tidal channels produce rapid facies transitions over short distances. This undermines the reliability of conventional deterministic seismic inversion methods, which struggle to resolve sub-seismic-scale layering.
- Limitations of existing inversion techniques, sparse-spike inversion suffers from limited vertical resolution, while geostatistical inversion is heavily dependent on dense well control and suffers from strong model dependency and lateral randomness. These limitations motivate the adoption of a more robust and data-integrated approach such as full-frequency inversion (FFI).

The key advantages of Waveform-driven full-frequency inversion (FFI) are the full utilization of 3D seismic waveform features, aligning better with geological trends, the integration of the geological concept of "facies control" into inversion, shifting the process from random simulation toward deterministic prediction, and the reduced subjectivity and randomness, resulting in higher accuracy and reliability. FFI

is applied to predict tidal channel layers as thin as 2–3 meters, significantly improving vertical and lateral resolution of reservoir prediction.

Methodology: Full-Frequency Seismic Inversion (FFI)

Overview

Full-frequency inversion (FFI) is a seismic waveform-driven inversion method that aims to reconstruct a broadband impedance model by integrating multiple frequency bands — low, mid, high, and ultra-high frequencies — using a structured spectrum synthesis approach. The method combines seismic waveform analysis, statistical clustering, and Bayesian inversion frameworks to provide high-resolution predictions of subsurface properties. As Figure 1 shows the seismic waveform is compared and analyzed with the well log bypass seismic track, the wells with similar waveform characteristics are selected as effective samples to construct a likelihood function that conforms to the seismic medium-frequency impedance and well curve structure characteristics.

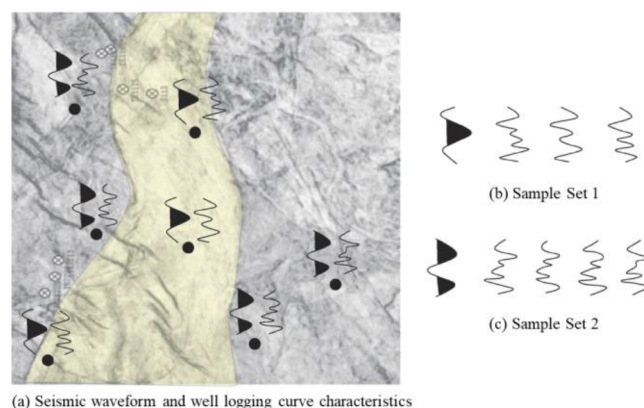


Figure 1—Basic schematic diagram of seismic waveform driven inversion

Building on traditional waveform indicator inversion techniques, FFI broadens the inversion bandwidth and overcomes limitations of conventional impedance inversion. It leverages structural characteristics captured by seismic waveforms and allows for integrated well-seismic inversion. FFI provides high precision in thin-bed prediction with strong reliability.

Using the "seismic waveform-driven structured statistical characterization" method, FFI models spatial variability in reservoir architecture by considering both waveform lateral variation and spatial distance. This is an improvement over conventional geostatistical methods that rely solely on spatial variograms.

The core principle of seismic waveform-driven full-frequency inversion lies in characterizing the spatial heterogeneity of reservoirs through the lateral variation of seismic waveforms. This technique interprets the seismic waveform as a composite response of multiple thin layers, thereby capturing key structural and compositional features of the reservoir. While individual thin layers are often beyond the resolution of conventional seismic data, variations in waveform shape can reflect underlying changes in reservoir structure and composition.

Building on this concept, representative wells are selected to identify a common structural frequency band, which serves as the deterministic component for constructing the initial model. Within a Bayesian inversion framework, high-frequency components of the initial model are simulated to align with both mid-frequency seismic impedance and the structural characteristics derived from well logs.

For the high-frequency domain, a waveform indicator inversion method — also referred to as facies-controlled interpolation — is employed. Well samples are classified based on waveform similarity, and a multi-scale analysis is conducted within each class. Deterministic structural components beyond the seismic

frequency band are extracted as facies-controlled simulation results, enabling a gradual transition from random to deterministic modeling of high-frequency components.

The inversion of high-frequency components is driven by the Sequential Monte Carlo Markov Chain (SMC-MC) algorithm, which serves as the core of waveform indicator inversion. This algorithm optimizes inversion by evaluating all wells based on spatial proximity and waveform similarity, assigning correlation scores to guide the selection of initial models. Wells with higher correlation yield inverted waveforms that more closely resemble the original seismic data, improving model reliability.

Two critical parameters in the SMC-MC algorithm are the effective sample size (n) and the optimal cutoff frequency. The effective sample size determines the number of wells contributing to each prediction point. A larger sample size lowers the waveform similarity threshold, resulting in smoother spatial continuity but reduced sensitivity to local heterogeneity. In contrast, a smaller sample size increases the similarity threshold, capturing more detailed spatial variations and enhancing heterogeneity in the inversion result.

The optimal cutoff frequency, set following the selection of the effective sample size, governs the frequency range included in the inversion. Controlled by co-depositional structural patterns, high-frequency inversion becomes increasingly random with higher frequencies. While increasing the cutoff frequency improves resolution, it also introduces greater stochasticity into the results.

By leveraging a global optimization strategy, the waveform indicator inversion approach transitions inversion results from fully stochastic to increasingly deterministic. Unlike conventional methods, this technique employs a non-spatial-domain statistical framework, making it robust to irregular well distribution—an inherent advantage in real-world applications. This makes it particularly well-suited for rolling exploration and development, especially in predicting thin reservoirs with significant lithological variability.

Frequency Band Composition

The FFI spectrum comprises four frequency bands, as [Figure 2](#) show schematic diagram of FFI principle:

- Low-Frequency (0–10 Hz): Derived from well logs and low-frequency models, capturing the overall structural trend and slow lateral variations. Velocity models contribute to constructing the low-frequency model with more reasonable spatial variability;
- Mid-Frequency (10–80 Hz): Extracted from seismic data, providing relative impedance information through relative acoustic impedance;
- High-Frequency (80–120 Hz): Estimated using waveform-facies-controlled inversion, which extend the inversion bandwidth and increase result determinism;
- Ultra-High-Frequency (>120 Hz): Estimated by combining well-log curves with seismic waveform using advanced intelligent cloud simulation technologies to improve well-log matching accuracy and prediction precision. incorporating waveform similarity and well-log statistics to estimate high-resolution attributes. The key challenge lies in ensuring that the simulated high-frequency content between wells is both realistic and geologically reasonable.

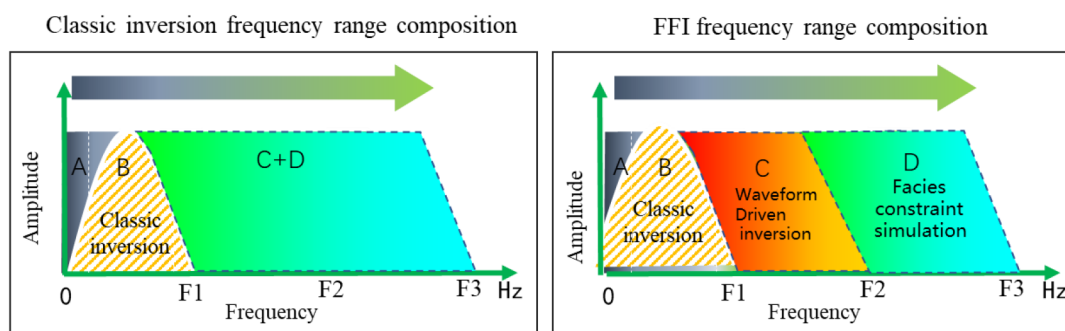


Figure 2—Schematic diagram of FFI principle

Shaped Spectrum Integration

The final full-frequency impedance volume is calculated using a shaped spectrum synthesis:

$$\text{FFI} = a * m * (F1) + b * n * (F2) + c * j * (F3) + d * k * (F4)$$

Where:

- FFI is the full-frequency composite spectrum;
- F1, F2, F3, F4 are the inversion results from the four frequency bands;
- a, b, c, d are weighting coefficients;
- m, n, j, k are spectrum shaping operators (e.g., triangular filters).

Waveform Clustering and Facies Control

To capture geological consistency, FFI uses dynamic clustering of seismic waveforms and correlates them with high-resolution well-log patterns. Wells with similar waveform and spatial proximity are selected to build statistical models that estimate high-frequency components. This facies-controlled approach ensures that inversion results honor both seismic data and geological trends.

FFI is a high-resolution seismic inversion technique driven by seismic waveform and integrated with well-log curves. Although seismic data are band-limited, seismic waveform lateral variations reflect depositional environment changes. Similar depositional environments typically share analogous sedimentary architecture, closely linked with waveform patterns. This enables estimation of high-frequency components by analyzing lateral waveform variations.

Building on this theoretical framework, the method incorporates both waveform similarity and spatial distance in statistical modeling. Wells with high correlation to the prediction point are prioritized for initial modeling to estimate high-frequency content optimally and unbiasedly. The final inversion result maintains consistency with the original seismic waveform, reflects seismic facies constraints spatially, and aligns with depositional geological patterns in the horizontal plane.

Application to Carbonate Tidal Channel Reservoirs

The FFI method was applied to predict the internal structure of tidal channel reservoirs in the A units of the target formation.

Petrophysical Classification

Based on porosity, permeability, and electrical properties, three lithofacies were identified:

- High-permeability tidal channel reservoirs, permeability >100mD and;
- Medium- to low-permeability tidal channel reservoirs, ;

- Non-reservoir facies.

Petrophysical analysis revealed that the dominant pore types were moldic and intergranular pores. Log responses, especially natural gamma, displayed upward-thinning cycles, with the best reservoirs concentrated in the lower part of the A unit. Figure 3 Seismic-well log facies classification based on porosity, permeability, and electrical properties.

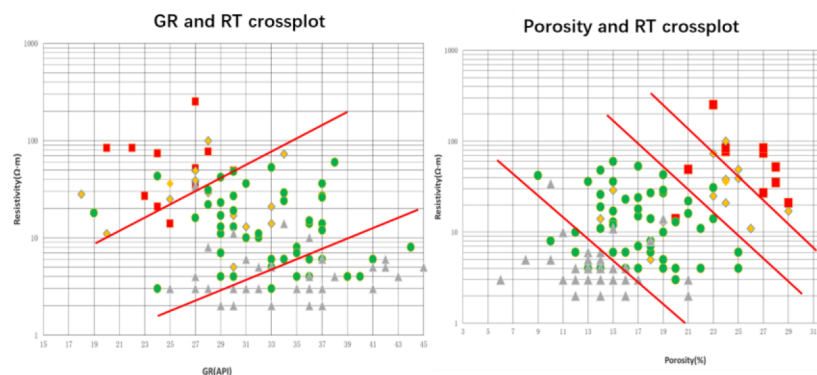


Figure 3—Lithfacies type of rock type, RT1: permeability >100mD red points and permeability 10~100mD, orange points; RT2: permeability 1~10mD green points, RT3: permeability less than 1mD grey points

Figure 4 (tidal channel distribution and inversion sections) shows that reservoir porosity in tidal channel sections is dominated by intergranular dissolution pores and moldic pores. The natural gamma log increases upward, forming thinning-upward cycles, with reservoirs mainly developed in the lower portion. Seismically, the tidal channels appear concave and band-shaped in profile, and grid-like in map view.

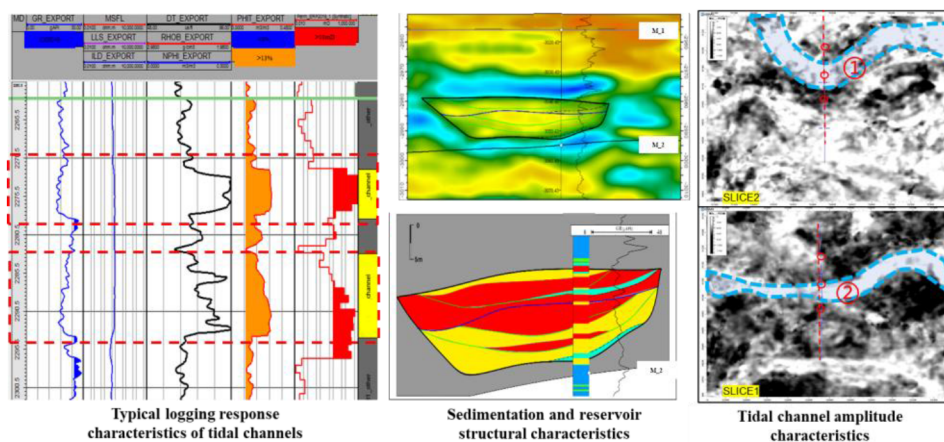


Figure 4—Tidal channel logging response, reservoir structure and amplitude characteristics

Inversion workflow and results

In the process of applying the seismic waveform-driven full-frequency inversion method for reservoir prediction, well log standardization, wavelet selection, and the choice of waveform indicator inversion parameters are the key aspects for achieving successful full-frequency inversion. FFI workflow is described as follows:

Step 1. Well Log Preprocessing, including curve standardization, reconstruction, and analysis of sensitive log responses.

In the process of oilfield exploration and development, it is difficult to ensure that all acquired well logging data are obtained using the same instruments and standardized calibration. As a result, the value

ranges of the same type of logging curves from different wells are often inconsistent. To accurately reflect the geological information of the strata, it is necessary to standardize the logging data.

The key to the standardization process lies in the selection of a reference (standard) well. Well data from the study area indicate that the target interval mainly consists of limestone. The sedimentation of the target formation is stable and well-layered. Notably, there is a 20-meter-thick layer of relatively pure mudstone above the target formation, which can serve as a marker bed for curve standardization.

Step 2. Framework Modeling, a sequence stratigraphic framework is constructed, followed by detailed characterization of depositional features and facies.

Select 3D seismic data and use the target horizon along with its overlying and underlying formation tops and bottoms, interpreted from seismic horizons, to construct a framework model. The selection of seismic interpretation horizons is based on the principle of not omitting any unconformities. There are typically three types of stratigraphic contacts between adjacent interpreted horizons and their overlying/underlying formations:

1. The overlying formation truncates the lower one;
2. The formations are parallel to each other;
3. The upper formation onlaps onto the lower one.

Accurately determine these stratigraphic relationships to create the stratigraphic framework model. In this study, four interpreted seismic horizons—A, B, C, and D—are selected from the 3D seismic data to construct the geological framework model, where all layers are assumed to be in parallel relationships.

Step 3. Model Interpolation, using the framework model as a constraint and applying the co-kriging algorithm, perform interpolation of the acoustic impedance curves from all wells within the work area, based on accurately calibrated seismic-geological data. This yields an acoustic impedance interpolation model for the target horizon across the work area. A low-frequency model data volume is obtained by using the seismic stacking velocity as a constraint.

Step 4. Spectral Simulation Inversion, extract the acoustic impedance curves between the top of horizon A and the bottom of horizon D for all wells, along with the corresponding seismic spectrum for that interval. Compare the well log spectrum with the seismic spectrum to determine a correction factor, which is then applied to seismic data for acoustic impedance inversion. This results in a relative impedance volume and a mid-frequency data volume.

Step 5. Waveform Indicator Inversion, using the framework model as a constraint and seismic data volume along with well acoustic impedance curves as input, perform statistical analysis. Through quality control, identify the optimal effective samples and cutoff frequencies for the selected wells. Then carry out waveform indicator inversion to obtain the high-frequency band volume.

In this step, seismic waveform attributes and well log data are analyzed jointly. Based on waveform similarity and spatial proximity, a set of reference wells is selected—those exhibiting high waveform similarity to the target trace. These wells are used to construct the initial model, and their P-wave impedance logs are statistically analyzed to define a prior probability distribution.

This initial model is then matched to the seismic impedance volume through filtering to derive the likelihood function. Multi-scale decomposition is applied to progressively remove high-frequency noise components. Based on Bayesian inference, the posterior probability distribution is obtained by combining the prior and likelihood, which serves as the objective function for inversion.

Since the waveform-selected samples exhibit strong spatial correlation, the Metropolis–Hastings algorithm is employed to sample from the posterior distribution. The realization corresponding to the maximum value of the objective function is identified as a feasible stochastic solution. The mean of multiple realizations is ultimately computed and used as the expected inversion output.

Step 6. Ultra-High-Frequency Waveform Simulation, based on the waveform inversion results, use seismic waveform as the driver and apply intelligent cloud simulation technology to enhance the fit with well data. This improves prediction accuracy and both vertical and lateral resolution.

Step 7. Frequency-Domain optimization merge of Volumes, using the optimal effective samples and cutoff frequencies, merge the inversion results from the four stages in the frequency domain according to their respective advantageous frequency bands. This produces the final inverted acoustic impedance result. The final full-frequency impedance volume is calculated using a shaped spectrum synthesis after optimized frequency domain merging.

Seismic sections revealed concave, ribbon-like features typical of tidal channels, while map-view amplitude slices exhibited net-like geometries. FFI inversion results showed clear low-impedance zones corresponding to high-quality reservoir intervals. These results aligned well with known well data, confirming the method's reliability.

Figure 5 shows ultra-high-resolution impedance sections showing channel stacking and lateral connectivity. Figure 5a shows the seismic amplitude characteristics of tidal channels in map view, with yellow, brown, and red indicating different tidal channels. Using FFI, the distribution of composite tidal channels was predicted. The results show that tidal channel reservoirs are clearly identifiable. They appear as low-amplitude features in seismic profiles (Figure 5b) and low-impedance features in inversion sections (Figure 5c).

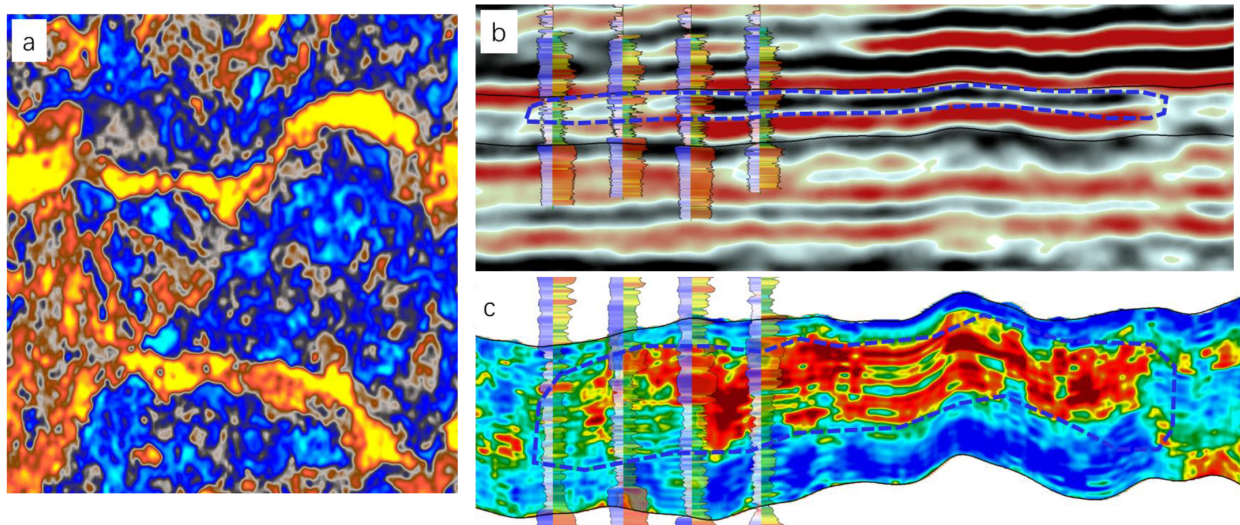


Figure 5—**a.** Tidal channel characteristics (yellow, brown and red represent tidal channel); **b.** Seismic profile (circle tidal channel); **c.** FFI prediction tidal channel (circle tidal channel, the red and yellow is tidal channel reservoir, blue is no-reservoir)

From the well-to-well acoustic impedance section (Figure 5), it can be observed that the acoustic impedance curves at the well points match well with the nearby traces. The vertical resolution of the acoustic impedance volume is relatively high. The inversion profile reveals that full-frequency inversion can clearly predict the vertical thin layers within tidal channels and delineate the lateral boundary features of the tidal channels.

Attribute Co-Analysis

FFI also enabled high-resolution simulation of non-impedance attributes such as resistivity and natural gamma. In the study area, FFI enables facies-controlled high-resolution resistivity simulation. In a tidal channel unit, high-permeability reservoir zones exhibit high porosity, high permeability, and high resistivity. FFI allows high-resolution facies-controlled simulation not only for acoustic impedance but also for other

non-impedance parameters such as natural gamma, resistivity, and porosity. This breaks the limitation of conventional seismic inversion, which typically yields only acoustic impedance.

Since high-permeability zones are critical for waterflood development in carbonate reservoirs, high-resolution resistivity simulation was used to identify these zones. By integrating acoustic impedance and resistivity volumes, high-resolution predictions of high-permeability reservoirs in units A were achieved. Additionally, integration of impedance and gamma ray volumes allowed a precise prediction of non-reservoir zones.

The co-rendering of wave impedance and resistivity volumes allowed clear identification of high-permeability thief zones, which are critical for optimizing waterflood design. Non-reservoir intervals were also identified by cross-plotting gamma and impedance volumes.

Results and Discussion

The FFI method improved vertical resolution to approximately 2 meters, enabling the identification of thin tidal layers and interbedded shales previously unresolved by conventional methods.

Lateral resolution was significantly enhanced through waveform similarity modeling, overcoming the randomness and over-reliance on prior models typical of geostatistical inversion.

Validation against both participating and independent wells showed >90% and >80% accuracy, respectively.

The integrated prediction of resistivity and porosity supported a more robust facies classification and sweet spot identification.

From a development perspective, the accurate delineation of high-permeability thief zones can directly inform infill well placement and layered water injection strategies, improving sweep efficiency and recovery.

Conclusions

- Full-frequency seismic inversion (FFI) leverages waveform-driven clustering and facies-controlled modeling to achieve high-resolution impedance prediction. It provides a robust solution to thin-layer identification in complex carbonate settings.
- FFI improves both vertical and lateral resolution beyond conventional methods. Its reliance on waveform features and geologically constrained clustering enables realistic simulation of thin reservoir architecture.
- Application in a northern UAE carbonate field demonstrated that FFI could accurately map tidal channel networks, interbeds, and high-permeability zones, supporting more effective reservoir management and development planning.

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